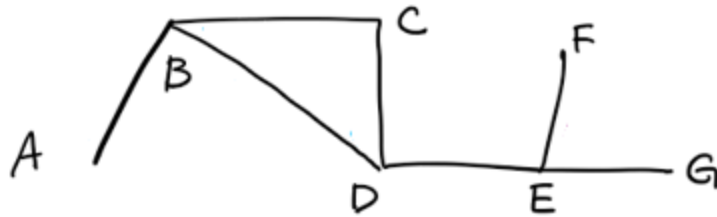


Lesson-2: Quiz (graded)

1

Multiple choice 1 point

Which of the following statements about this graph is **False**?

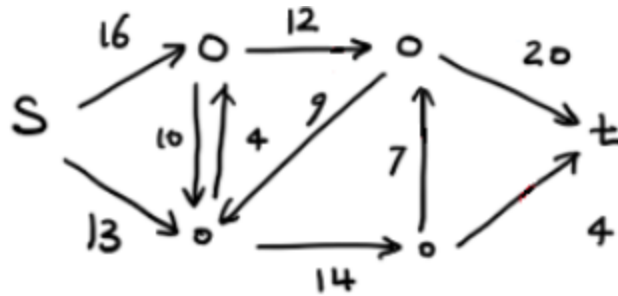


- ☐ The sequence 'BDCB' is a walk and a cycle
- ☐ The sequence 'BCDBA' is a path
- ☐ The sequence 'GEF' is a walk
- ☐ The sequence 'FEG' is a path

2

Multiple choice 1 point

Find the max flow of the following network

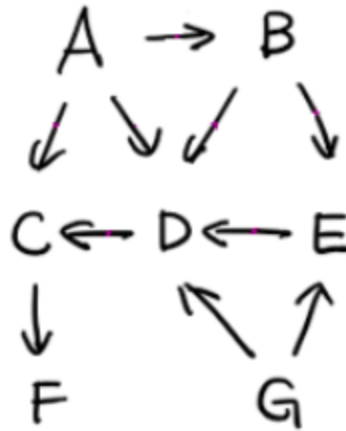


- ☐ 20
- ☐ 23
- ☐ 21
- ☐ 24

3

Multiple choice 1 point

Which of the following node sequences is **NOT** a topological order of this graph?



- ☐ AGBEDCF
- ☐ GABEDCF
- ☐ ABGEDCF
- ☐ ABEDGCF

4

Multiple choice 1 point

Which of the following statements is **False**

- ☐ In a flow network, the amount of maximum flow is always equal to capacity of the minimum cut.
- ☐ An undirected graph is connected if and only if there exist(s) one or more paths from any node to any other node
- ☐ A graph is a DAG if and only if there is a topological ordering of its nodes
- ☐ Two nodes, s and t, belong in the same strongly connected component if and only if there is a path from s to t

5

Multiple choice 1 point

Which of the following expressions is equal to the number of walks of length 2 between two distinct nodes i, j in a graph with adjacency matrix A ? You can assume that there are no self-loops.

1) $A_{i,j}$ (where $i \neq j$)

2) $\sum_{k=1}^n A_{i,k} A_{k,j}^T$ (where $i \neq j$, $j \neq k$, and $k \neq i$)

3) $\sum_{k=1}^n A_{i,k} A_{k,j}$ (where $i \neq j$, $j \neq k$, and $k \neq i$)

4) None of the above

☐ 1

☐ 2

☐ 3

☐ 4

6

Multiple choice 1 point

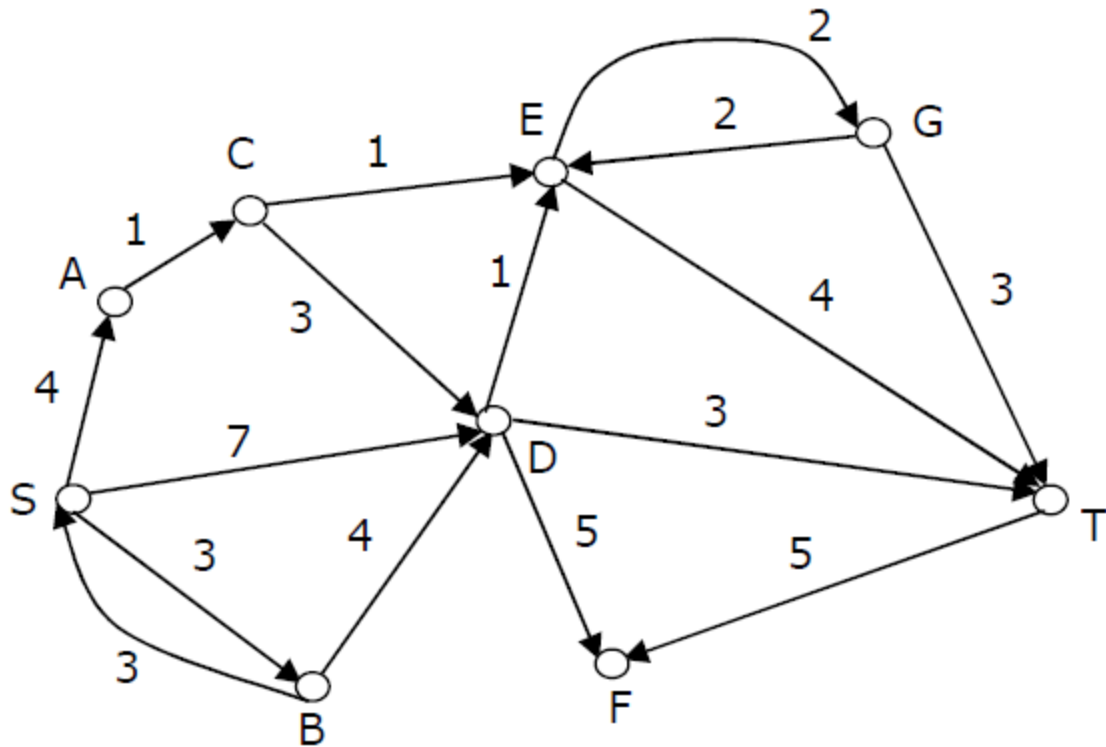
Which of the following statements about an Adjacency matrix and Adjacency list is **True**?

- ☐ The size of the Adjacency matrix is $n \times m$ where n is the number of vertices and m is the number of edges
- ☐ Most elements of the adjacency matrix of a dense graph are zeros.
- ☐ The number of adjacency list's entries is n plus the sum of all node degrees.
- ☐ A graph is sparse only if the number of edges remains constant as the network grows.

7

Multiple choice 1 point

Consider the directed graph shown in the figure below. There are multiple shortest paths between vertices S and T. Which one will be reported by Dijkstra's shortest path algorithm? Assume that, in any iteration, the shortest path to a vertex v is updated only when a strictly shorter path to v is discovered



- ☐ SDT
- ☐ SBDT
- ☐ SACDT
- ☐ SACET